Anna Stuhlmacher

Department of Electrical and Computer Engineering
Michigan Technological University - Houghton, MI

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Education

University of Michigan — **Ann Arbor, MI**
Ph.D. - Electrical Engineering — *2023*
Advisor: Johanna L. Mathieu

University of Michigan — **Ann Arbor, MI**
M.S. - Electrical Engineering — *2019*

Boston University — **Boston, MA**
B.S. - Electrical Engineering — *2017*
Summa Cum Laude

Positions

Michigan Technological University — **Houghton, MI**
Assistant Professor, Electrical and Computer Engineering — *2023-Present*
Affiliated Assistant Professor, Mechanical and Aerospace Engineering — *2026-Present*
Affiliated Assistant Professor, Civil, Environmental, and Geospatial Engineering — *2026-Present*

University of Michigan — **Ann Arbor, MI**
Graduate Student Research Assistant — *2017-2023*
Graduate Student Instructor — *Fall 2021*
Undergraduate Researcher, Summer Research Opportunity Program — *Summer 2016*

National Renewable Energy Laboratory* — **Golden, CO**
Power Systems Control and Optimization Intern, — *Summer 2021*
**Now called: National Laboratory of the Rockies*

Boston University — **Boston, MA**
Undergraduate Research Assistant in Joshua Semeter's Lab — *2014-2017*

Publications and Presentations

** indicates presenter, underline indicates students mentored*

Journal Papers

[J7] **A. Stuhlmacher***, P. Srisuthankul, J.L. Mathieu, P. Seiler, "A Model Predictive Control Approach to Dual-Axis Agrivoltaic Panel Tracking", In: Renewable Energy, vol. 274, p. 126155. October 2026. DOI: 10.1016/j.renene.2026.126155.

[J6] **A. Stuhlmacher***, A. Kody, and M. Wu, "Optimizing Biogas Use in Wastewater Treatment Plants For Demand Flexibility", In: Sustainable Energy, Grids and Networks - Special Issue for the 2025 IREP Symposium on Bulk Power System Dynamics and Control, vol. 43, p. 101825. Sorrento, Italy, September 2025. DOI: 10.1016/j.segan.2025.101825.

[J5] **A. Stuhlmacher**, S. Guikema, and J. L. Mathieu, "Assessing Power and Water Network Resilience When Water Pumps Provide Frequency Regulation", In: IEEE Transactions on Power Systems, vol. 40, no. 5, pp. 3833 - 3845. September 2025. DOI: 10.1109/TPWRS.2025.3539288.

[J4] **A. Stuhlmacher**, C. Ten, L. Dilworth, and Y. Tang, "Operational Planning for Emerging Distribution Systems: A Unique Perspective on Grid Expansion", In: Foundations and Trends in Electric Energy Systems, vol. 7, no. 2, pp. 63-164. December 2023. DOI: 10.1561/3100000033.

[J3] **A. Stuhlmacher*** and J. L. Mathieu, "Flexible Drinking Water Pumping to Provide Multiple Grid Services", In: Electric Power Systems Research - Special Issue for the 2022 Power Systems Computation Conference (PSCC), vol. 212, p. 108491. Porto, Portugal, June 2022. DOI: 10.1016/j.epsr.2022.108491.

[J2] **A. Stuhlmacher** and J. L. Mathieu, "Chance-Constrained Water Pumping to Manage Water and Power Demand Uncertainty in Distribution Networks," In: Proceedings of the IEEE, vol. 108, no. 9, pp. 1640-1655. September 2020. DOI: 10.1109/JPROC.2020.2997520.

[J1] **A. Stuhlmacher*** and J. L. Mathieu, "Water Distribution Networks as Flexible Loads: A Chance-constrained Programming Approach", In: Electric Power Systems Research - Special Issue for the 2020 Power Systems Computation Conference (PSCC), vol. 188, p. 106570, (virtual), June 2020. DOI: 10.1016/j.epsr.2020.106570. *Presentation Link.*

Conference Proceedings

[C12] M. Wu* and **A. Stuhlmacher**, "Optimal Sizing and Dispatch of Biogas Storage and CHP for Demand Response in Wastewater Treatment Plants", In: (Under Review).

[C11] G. E. Myers* and **A. Stuhlmacher**, "Three-Phase Optimal Operation of Soft Open Points with Integrated Energy Storage", In: (Under Review).

[C10] A. N. Sakib*, C. Crozier, and **A. Stuhlmacher**, "Assessing the Impact of Electricity Tariff Design on Water Distribution Network Flexibility", In: (Under Review).

[C9] M. Wu* and **A. Stuhlmacher**, "Productive Curtailment in Agrivoltaic Systems under Flexible Interconnection Agreements", In: PowerUp 2026 Conference. Boulder, Colorado, September 2026. DOI: 10.48550/arXiv.2607.10040.

[C8] **A. Stuhlmacher*** and A. Kody, "Multi-Stage Robust Optimization of a Wastewater Treatment Biogas Generator Participating in Day-Ahead and Real-Time Regulation Markets", In: Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS). Lahaina, Hawaii, January 2026. DOI: 10125/111771.

[C7] M. Wu* and **A. Stuhlmacher**, "Investigating Agrivoltaic System Design Choices on Shading and Power Production", In: Proceedings of the North American Power Symposium (NAPS). Hartford, Connecticut, October 2025. DOI: 10.1109/NAPS66256.2025.11272386.

[C6] A. N. Sakib* and **A. Stuhlmacher**, "Leveraging Drinking Water Pumps as Flexible Loads Using Input Convex Neural Networks", In: Proceedings of the IEEE Power and Energy Society General Meeting (PES GM). Austin, Texas, July 2025. DOI: 10.1109/PESGM52009.2025.11225422.

[C5] **A. Stuhlmacher*** and J. L. Mathieu, "Demand Response Potential of Drinking Water Distribution Networks", In: Proceedings of the 58th Hawaii International Conference on System Sciences (HICSS). Waikoloa Village, Hawaii, January 2025. DOI: 10125/109192.

[C4] **A. Stuhlmacher***, J. L. Mathieu, and P. Seiler, "Optimizing Dual-Axis Solar Panel Operation in an Agrivoltaic System and Implications for Power Systems", In: Proceedings of the 57th Hawaii International Conference on System Sciences (HICSS). Waikiki, Hawaii, January 2024. DOI: 10125/106735.

[C3] **A. Stuhlmacher*** and J. L. Mathieu, "Uncertainty-Aware Methods for Leveraging Water Pumping Flexibility for Power Networks", In: Proceedings of the IREP Symposium on Bulk Power System Dynamics and Control. Banff, Canada, August 2022. DOI: 10.48550/arXiv.2207.04943.

[C2] **A. Stuhlmacher***, L. A. Roald, and J. L. Mathieu, "Tractable Robust Drinking Water Pumping to Provide Power Network Voltage Support", In: Proceedings of the Conference on Decision and Control (CDC), pp. 4206-4213. (virtual), December 2021. DOI: 10.1109/CDC45484.2021.9683419.

[C1] **A. Stuhlmacher*** and J. L. Mathieu, "Chance-Constrained Water Pumping Managing Power Distribution Network Constraints", In: Proceedings of the North American Power Symposium (NAPS). Wichita, KS, October 2019. DOI: 10.1109/naps46351.2019.9000282.

Dissertation

A. Stuhlmacher, "Optimal Scheduling and Control of Uncertain Coupled Power-Water Distribution Networks". PhD Thesis. University of Michigan. May 2023. DOI: 10.7302/7426.

Technical Reports

[T1] R. O'Neil, K. Oikonomou, M. Parvania, V. Tidwell, A. T. Al-Awami, M. Panteli, S. Conrad, T. Brekken, E. Goharian, N. Voisin, "Integrated Water and Power Systems: Current State and Research Roadmap," IEEE PES Task Force on Water-Power Systems, Technical Report No. PES-TR114, September 2023. *Contributor to the 'Integrated Operation of Water and Power Systems' Topic Area*

Abstracts with Oral Presentations

[A3] **A. Stuhlmacher***, S. Guikema, and J. L. Mathieu, "Assessing the Resilience of an Optimal Water Pumping Control Strategy to Provide Frequency Regulation", INFORMS Annual Meeting. Phoenix, AZ, October 2023.

[A2] **A. Stuhlmacher*** and J. L. Mathieu, "Stochastic Optimization of Water Distribution Network Operation to Provide Power Grid Flexibility", SIAM Conference on Optimization Annual Meeting. Seattle, WA, May 2023.

[A1] **A. Stuhlmacher***, L. A. Roald, and J. L. Mathieu, "An Adjustable Robust Optimization Model for Drinking Water Pumping as a Flexible Load", INFORMS Annual Meeting. (virtual), October 2021.

Posters

- [P18] M. Wu* and **A. Stuhlmacher**, "Optimal Sizing and Dispatch of Biogas Storage and Combined Heat and Power Systems for Demand Response in Wastewater Treatment Plants", Alumni Reunion Poster Session. Houghton, MI, July 2026.
- [P17] A. N. Sakib* and **A. Stuhlmacher**, "Assessing the Impact of Electricity Tariff Design on Water Distribution Network Flexibility", Alumni Reunion Poster Session. Houghton, MI, July 2026.
- [P16] A. N. Sakib* and **A. Stuhlmacher**, "Assessing the Impact of Electricity Tariff Design on Water Distribution Network Flexibility", Research Universities for Michigan Water Security Workshop. Houghton, MI, July 2026.
- [P15] C. Carrillo* and **A. Stuhlmacher**, "Modeling Electric Heating Load and Flexibility in Cold Climate Homes Using the OCHRE Model", Undergraduate Research Symposium. Houghton, MI, June 2025.
- [P14] A. N. Sakib* and **A. Stuhlmacher**, "Leveraging Drinking Water Pumps as Flexible Loads Using Input Convex Neural Networks", Graduate Research Colloquium. Houghton, MI, March 2025.
- [P13] M. Wu* and **A. Stuhlmacher**, "Investigating the Impact of Agrivoltaic Design Choices on Crop Shading and Electricity Production", Undergraduate Research & Scholarship Symposium. Houghton, MI, March 2025.
- [P12] A. N. Sakib* and **A. Stuhlmacher**, "Flexible Operation of Drinking Water Pumps using Input Convex Neural Network Approximations", Institute of Computing and Cybersystem's Computing Showcase. Houghton, MI, October 2024. (2nd place within Graduate Student Category)
- [P11] M. Wu* and **A. Stuhlmacher**, "Investigating the Impacts of Agrivoltaic Design Choices on Inter-Row Shading and Electricity Production", Institute of Computing and Cybersystem's Computing Showcase. Houghton, MI, October 2024. (2nd place within Undergraduate Student Category, selected for publication in the College of Computing's undergraduate journal, *Infinite Loop*)
- [P10] **A. Stuhlmacher*** and J. L. Mathieu, "Assessing the Resilience of an Optimal Water Pumping Strategy to Provide Frequency Regulation", IEEE Power and Energy Society General Meeting. Orlando, FL, July 2023.
- [P9] **A. Stuhlmacher***, J. L. Mathieu, and P. Seiler, "Optimizing Dual-Axis Solar Panel Operation in an Agrivoltaic System under Uncertainty", AgriVoltaics2023 Conference and Exhibition. (virtual), April 2023. *Presentation Link.*
- [P8] **A. Stuhlmacher*** and J. L. Mathieu, "Computationally Tractable Uncertainty-Aware Framework for Optimal Water Pumping in Coupled Power-Water Systems", Fifth Workshop on Autonomous Energy Systems, National Renewable Energy Laboratory (NREL). Golden, CO, July 2022.
- [P7] D. Li*, **A. Stuhlmacher**, and J. L. Mathieu, "Estimating the Demand Response Potential of Drinking Water Distribution Networks in Arizona", University of Michigan Undergraduate Research Symposium. Ann Arbor, MI, April 2022.
- [P6] C. Bertcher*, **A. Stuhlmacher**, and J. L. Mathieu, "Comparison of Linearized Three-Phase Unbalanced Power Flow Models", IEEE Power and Energy Society General Meeting. (virtual), July 2021. *Presentation Link.*
- [P5] C. Bertcher*, **A. Stuhlmacher**, and J. L. Mathieu, "UM Bus Electrification: Challenges and Solutions", University of Michigan Undergraduate Research Symposium. Ann Arbor, MI, April 2019.
- [P4] **A. Stuhlmacher*** and J. L. Mathieu, "Stochastic Water Distribution Network Operation Considering Power Distribution Network Constraints", Engineering Graduate Symposium, University of Michigan. Ann Arbor, MI, October 2018.
- [P3] **A. Stuhlmacher***, J. L. Mathieu, and V. Gupta, "Water-Power Distribution Network Coupling for Optimal Pumping to Reduce Energy Costs and Promote Resilience", Engineering Graduate Symposium, University of Michigan. Ann Arbor, MI, November 2017.
- [P2] **A. Stuhlmacher***, S. Crocker, and J. L. Mathieu, "Effects of Aggregate Load Control on the Physical Constraints of Distribution Networks", Rackham Summer Research Opportunity Program Symposium, University of Michigan. Ann Arbor, MI, July 2016.
- [P1] S. Crocker*, **A. Stuhlmacher**, and J. L. Mathieu, "Effects of Aggregate Load Control on the Physical Components of Distribution Networks", IEEE PES General Meeting. Boston, MA, July 2016.

Honors and Awards

- Professor of the Year, IEEE HKN Michigan Tech Chapter, 2025-2026
- Selected by the College of Engineering Dean for the Deans' Teaching Showcase, Michigan Technological University, Spring 2026 (Link)

- Best paper award for the Electric Energy Systems Track, Hawaii International Conference on System Sciences, January 2025
- Societal Impact Award, Senior Design Capstone Project, College of Engineering, Boston University, Spring 2017
- Entrepreneurial Award, Senior Design Capstone Project, Department of Electrical and Computer Engineering, Boston University, Spring 2017

Funding

REU Supplement to: ERI: Optimal Demand Response Strategies Using Biogas from Wastewater Treatment **$10,000**
National Science Foundation (NSF)
July 2026-June 2027
PI

Advancing Coordination of Drinking Water and Electric Power Systems for Sustainable Michigan Infrastructure **$86,595**
Research Universities for Michigan (RU4M)
February 2026–January 2027
PI ($26,595), with co-PIs Maggie Williams (MSU), Annick Anctil (MSU), Johanna Mathieu (UM), Branko Kerkez (UM), Carol Miller (WSU), Caisheng Wang (WSU)

ERI: Optimal Demand Response Strategies Using Biogas from Wastewater Treatment **$200,000**
National Science Foundation (NSF)
July 2025-June 2027
PI

Stochastic Temperature-Dependent Models for Evaluating Flexible Load Dispatch **$70,000**
Power Systems Engineering Research Center (PSERC)
August 2025-July 2027
Total: $220,000
Co-PI, with PI Constance Crozier (Georgia Tech) and Co-PI Daniel Molzahn (Georgia Tech)

R2I2: Maximizing Resilience to Winter Weather in Future Electric Power Systems **$7,230**
National Science Foundation (NSF)
August 2025-July 2027
Total: $499,314
SP, with PI Ana Dyreson (Michigan Tech) and Co-PIs Chelsea Schelly (Michigan Tech), Pengfei Xue (Michigan Tech), and Vivek Srikrishnan (Cornell)

REF RS: Development and Validation of Single-Axis Solar Tracking Systems for Agrivoltaics **$36,340**
Research Excellence Fund - Research Seed Grant, Michigan Technological University
July 2025-June 2026
PI, with co-PI Ana Dyreson (Michigan Tech) and co-PI Chelsea Schelly (Michigan Tech)

Course Innovation Grant **$1,500**
IDEAhub, Michigan Technological University
June 2024-August 2024

Flexible Operation of Drinking Water Pumps Using Learning-Aided Optimization **$9,445**
Rapid Seedling Award, GLRC-ICC Joint Institute, Michigan Technological University
May 2024-August 2024
PI

Rackham Predoctoral Fellowship **$44,214**
Rackham Graduate School, University of Michigan
May 2022-April 2023

Graduate Research Fellowship Program (GRFP) **$138,000**
National Science Foundation
2017-2020

Invited Talks

- Michigan Technological University, Environmental Engineering Graduate Seminar, “Coordinating Drinking Water and Electric Power Systems to Support System Reliability”, March 23, 2026.
- Michigan Water Environment Association (MWEA) Administrators Workshop, “Strategies for Integrating Wastewater Plants into the Power Grid", December 2nd, 2025.
- Texas A&M, “Coordination of the Water and Power Sectors to Provide Grid Flexibility”, March 28th, 2025.
- IEEE Northeastern Wisconsin Section, “Coordination of the Water Supply System and the Power Grid to Support System Performance”, February 6th, 2025.
- Purdue University, Herrick Energy Seminar, “Coordination of the Water Supply System and the Power Grid to Support System Performance”, April 26th, 2024.
- IEEE Northeastern Wisconsin Section, “Agrivoltaics - Placing Solar Photovoltaic Panels Over Cropland”, February 15th, 2024.
- Polytechnique Montréal, Group for Research in Decision Analysis (GERAD), “Optimizing Dynamic Solar Panel Operation in an Agrivoltaic System and Implications for Power Systems” (virtual), January 24th, 2024.
- Michigan Technological University, Alternative Energy Enterprise, “Agrivoltaics - Placing Solar Photovoltaic Panels Over Cropland”, November 28th, 2023.
- Stanford University, Water and Energy Efficiency for the Environment Lab (WE3Lab), “Optimizing Flexible Drinking Water Networks to Support Power System Performance” (virtual), July 14th, 2023.
- Cornell University, “Optimizing Flexible Drinking Water Networks to Support Power System Performance”, March 13th, 2023.
- Oregon State University, “Optimizing Flexible Drinking Water Networks to Support Power System Performance”, February 22nd, 2023.
- Michigan Technological University, “Optimizing Flexible Drinking Water Networks to Support Power System Performance”, February 6th, 2023.
- Portland State University, “Optimizing Flexible Resources to Support Power System Resiliency”, January 11th, 2023.
- Hope College, “Drinking Water Networks as Flexible Loads in the Power Grid”, November 12th, 2021.

Teaching

MTU EE 4263/5263: Distributed Energy Resources — **Houghton, MI**
Instructor — *Fall ’25*

MTU EE 5232: Power System Optimization — **Houghton, MI**
Instructor — *Spring ’24, ’27*

MTU EE 3120: Electrical Energy Systems — **Houghton, MI**
Instructor — *Fall ’23, ’24, ’25, ’26*
Spring ’25, ’26

UM EECS 460: Control Systems Analysis and Design — **Ann Arbor, MI**
Graduate Student Instructor — *Fall ’21*

BU EC 402: Introduction to Control Systems — **Boston, MA**
Undergraduate Teaching Fellow — *Spring ’17*

Guest Lecture

- UM EECS 534: Analysis of Electric Power Distribution Grids and Loads,
  “Power Flow Relaxations and Approximations for Unbalanced Networks”, October 12th, 2022.

Graduate Teacher Certificate — **Ann Arbor, MI**
University of Michigan, Center for Research on Learning and Teaching (CRLT) — *Spring ’22*

UM EECS 598: Markets and Optimization — **Ann Arbor, MI**
Grader — *Fall ’19, Spring ’22*

Service

Society Memberships

** intermittently*

Institute of Electrical and Electronics Engineers (IEEE), IEEE Power and Energy Society (PES), Institute for Operations Research and the Management Sciences (INFORMS)*, Graduate Society of Women Engineers, Tau Beta Pi Engineering Honors Society, IEEE HKN Boston University Chapter, Order of the Engineer

Technical Committees

- IEEE PES Task Force on Water-Power Systems

Conferences, Workshops, and Panels

- Technical Program Committee member for the 2026 International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), September 2026.
- Technical Agenda Chair for the North American Power Symposium (NAPS), October 2026.
- Technical Program Committee member for the PowerUp 2026 Conference, September 2026
- Paper session chair and paper judge for the North American Power Symposium (NAPS), October 2025.
- Judge for the Student Poster Competition at the IEEE Power and Energy Society General Meeting (PES GM).
- International Technical Committee Member for IEEE PES Innovative Smart Grid Technologies (ISGT) Europe 2025 Conference.
- Panelist for "Tenure-Track Faculty Positions", University of Michigan ENGR 580 (Teaching Engineering), Dec. 2, 2024.
- Session organizer and chair of the Distributed, Renewable and Mobile Resources minitrack, Hawaii International Conference on System Sciences (HICSS), 2025 and 2026.

Reviewer

Journals

IEEE Transactions on Power Systems; IEEE Transactions on Control of Networked Systems; IEEE Transactions on Smart Grids; IEEE Transactions on Sustainable Energy; Electric Power Systems Research; IEEE Power Engineering Letters; IEEE Control Systems Letters (L-CSS); ASCE Journal of Water Resources Planning & Management; IEEE Transactions on Transportation Electrification; Sustainable Energy, Grids and Networks; Renewable and Sustainable Energy Reviews

Conferences

Power Systems Computation Conference (PSCC), IEEE Conference on Decision and Control (CDC), Conference on Probabilistic Methods Applied to Power Systems (PMAPS), American Control Conference (ACC), Hawaii International Conference on System Sciences (HICSS), IEEE PES Innovative Smart Grid Technologies (ISGT) Europe, North American Power Symposium (NAPS), PowerUp Conference

Proposals

National Science Foundation, Michigan Tech Research Excellence Fund

In The News

- “U-M faculty, partners awarded state water research grants", The University Record, February 2nd, 2026.
- “Michigans Research Universities Unite to Tackle States Toughest Water Challenges", Michigan RU4M, January 23rd, 2026.
- “MTU College of Engineering Debuts New Research Experience for Undergraduate Students”, Michigan Technological University Unscripted Research Blog, April 15th, 2025.
- “Johanna Mathieu and Anna Stuhlmacher receive HICSS Best Paper Award for work on the potential of drinking water networks as flexible electric loads”, University of Michigan ECE News, February 7th, 2025.
- “Holiday Lights Survey: When Do Americans Start Decorating?", This Old House, October 29, 2024.